Machine Learning Lab

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Lab 5

Part 1:

1.

```
library(data.table)
library(dplyr)
library(ggplot2)
library(slam)          # New: needed for perplexity calc
library(quanteda)      # New: to process text
library(topicmodels)   # New: to fit topic models
library(word2vec)      # New: to fit word embeddings
library(tidyr)
library(purrr)
library(knitr)
library(kableExtra)
```

```
fb_congress_data <- fread("~/Documents/ML/Labs/Lab5/fb-congress-data3.csv")
dim(fb_congress_data)
```

```
[1] 6752    4
```

```
colnames(fb_congress_data)
```

```
[1] "doc_id"       "screen_name" "party"       "message"
```

The dataset contains 6752 rows and 4 columns. The variable names are: `doc_id`, `screen_name`, `party`, and `message`.

2

```r
posts_corpus <- corpus(x = fb_congress_data,
                       text_field = "message",
                       meta = list("screen_name", "party"),
                       docid_field = 'doc_id')

# print(posts_corpus)

# Tokenize & clean from particular types of words
mytokens <- tokens(x = posts_corpus,
                   remove_punct = TRUE,
                   remove_numbers = TRUE,
                   remove_symbols = TRUE,
                   remove_url = TRUE,
                   padding = FALSE)
# print(mytokens)

# Remove English stop words
mytokens <-   tokens_remove(x = mytokens,
                            stopwords("en"),
                            padding = FALSE)
# Remove @'s
mytokens <-   tokens_select(x = mytokens,
                            selection = 'remove',
                            valuetype = 'glob',
                            pattern = '@',
                            padding = FALSE)

# Make tokens lowercase
mytokens <- tokens_tolower(x = mytokens)

# How does it look now?
mytokens[1:3]
```

```
Tokens consisting of 3 documents and 2 docvars.
1 :
 [1] "president"     "trump"         "backs"         "paris"
 [5] "agreement"     "economic"      "environmental" "national"
 [9] "security"      "moral"         "disaster"      "united"
[ ... and 6 more ]
```

```
2 :
 [1] "many"       "thanks"     "first"      "class"      "summer"
 [6] "interns"    "washington" "hard"       "work"       "folks"
[11] "#ga03"

3 :
 [1] "economy"    "needs"      "shot"       "arm"        "spur"
 [6] "growth"     "co-chair"   "bipartisan" "problem"    "solvers"
[11] "caucus"     "i've"
[ ... and 27 more ]
```

I printed the first three texts for inspection, and I can confirm that the data has been successfully processed into tokens.

```r
# Create document term matrix
dtm <- dfm(x = mytokens)

# Exclude words with too low frequency
dtm <- dfm_trim(dtm, min_termfreq = 5)

# Exclude documents with too low frequency
rowsums <- rowSums(dtm)
keep_ids <- which(rowsums>=10)
dtm <- dtm[keep_ids,]

# Report final dimensionality of data set
dim(dtm)
```

```
[1] 5748 5484
```

The final document-term matrix has dimensions of 5,748 rows (documents) and 5,484 columns (terms).

3. LDA

```r
# Fit LDA (package: topicmodels)
# - 30 topics
# - Estimation algorithm: Gibbs
# - 1000 iterations
```

```r
# - Set seed for replicability
# - Print iter every 1000th (I need the "quiet" model)
set.seed(1)
K <- 50
mylda <- LDA(x = dtm,
             k = K,
             method="Gibbs",
             control=list(iter = 1000,
                          seed = 1,
                          verbose = 0))
```

4.

```r
# Inspect top 15 from each topics
get_terms(mylda, 15)
```

```
      Topic 1     Topic 2        Topic 3          Topic 4     Topic 5
 [1,] "program"   "states"       "u.s"            "need"      "week"
 [2,] "families"  "united"       "north"          "help"      "last"
 [3,] "children"  "rights"       "military"       "resources" "night"
 [4,] "community" "amendment"    "korea"          "crisis"    "see"
 [5,] "services"  "people"       "strategy"       "opioid"    "also"
 [6,] "need"      "right"        "administration" "like"      "many"
 [7,] "also"      "justice"      "international"  "address"   "three"
 [8,] "grant"     "civil"        "region"         "get"       "washington"
 [9,] "seniors"   "constitution" "world"          "epidemic"  "able"
[10,] "centers"   "without"      "nuclear"        "drug"      "read"
[11,] "kids"      "second"       "foreign"        "treatment" "hear"
[12,] "provide"   "human"        "south"          "helping"   "time"
[13,] "start"     "individuals"  "policy"         "abuse"     "events"
[14,] "keep"      "equality"     "syria"          "combat"    "good"
[15,] "chip"      "americans"    "countries"      "21st"      "lot"
      Topic 6    Topic 7          Topic 8       Topic 9         Topic 10
 [1,] "state"    "president"      "americans"   "energy"        "committee"
 [2,] "today"    "trump"          "millions"    "change"        "house"
 [3,] "senator"  "administration" "bill"        "power"         "hearing"
 [4,] "home"     "donald"         "republicans" "climate"       "today"
 [5,] "see"      "trump's"        "million"     "clean"         "chairman"
 [6,] "west"     "trump's"        "coverage"    "future"        "member"
 [7,] "virginia" "order"          "health"      "protect"       "subcommittee"
```

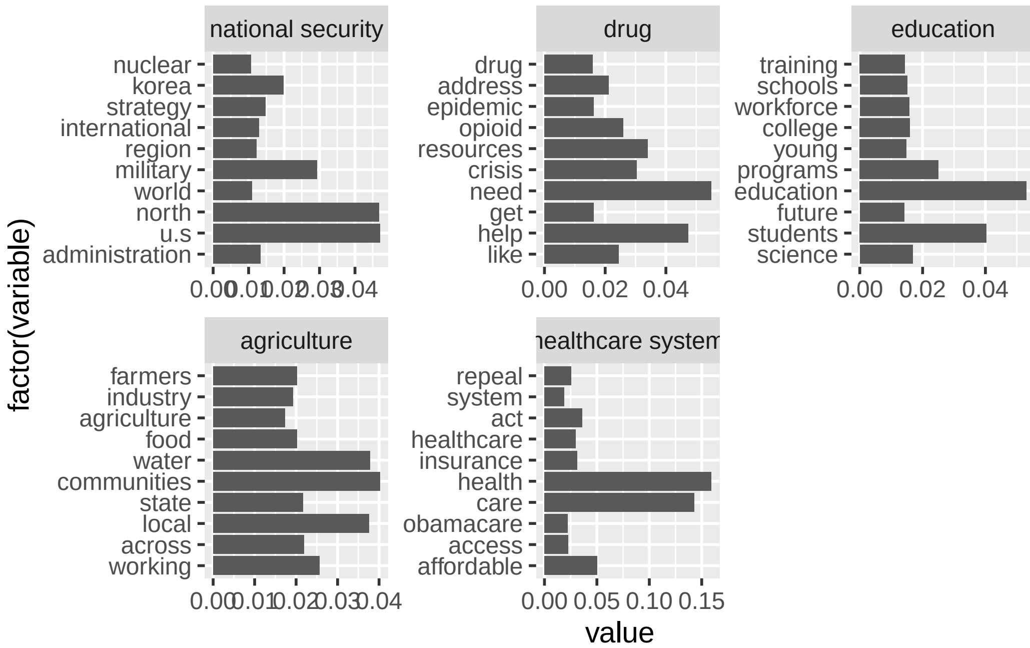
```
 [8,] "place"    "executive"      "people"      "environment"   "american"
 [9,] "across"   "j"              "plan"        "epa"           "want"
[10,] "read"     "now"            "medicaid"    "environmental" "oversight"
[11,] "glad"     "decision"       "away"        "back"          "said"
[12,] "take"     "actions"        "insurance"   "progress"      "leadership"
[13,] "hope"     "signed"         "trumpcare"   "understand"    "well"
[14,] "airport"  "action"         "care"        "commitment"    "including"
[15,] "grassley" "obama"          "conditions"  "generations"   "congresswoman"
      Topic 11 Topic 12       Topic 13     Topic 14     Topic 15
 [1,] "us"     "office"       "federal"    "life"       "general"
 [2,] "know"   "washington"   "assistance" "history"    "investigation"
 [3,] "like"   "district"     "help"       "month"      "director"
 [4,] "fight"  "staff"        "u.s"        "story"      "russia"
 [5,] "can"    "visit"        "puerto"     "first"      "former"
 [6,] "every"  "please"       "rico"       "read"       "russian"
 [7,] "want"   "dc"           "hurricane"  "made"       "sessions"
 [8,] "many"   "learn"        "disaster"   "black"      "intelligence"
 [9,] "one"    "constituents" "relief"     "celebrate"  "independent"
[10,] "give"   "d.c"          "recovery"   "stories"    "fbi"
[11,] "let"    "call"         "emergency"  "helped"     "must"
[12,] "even"   "visiting"     "support"    "take"       "attorney"
[13,] "share"  "tour"         "fema"       "came"       "special"
[14,] "mr"     "capitol"      "efforts"    "shared"     "election"
[15,] "ever"   "thank"        "provide"    "experience" "democracy"
      Topic 16     Topic 17      Topic 18      Topic 19      Topic 20
 [1,] "education"  "law"         "system"      "great"       "honor"
 [2,] "students"   "keep"        "country"     "meeting"     "court"
 [3,] "programs"   "safe"        "immigration" "issues"      "war"
 [4,] "science"    "enforcement" "daca"        "yesterday"   "honored"
 [5,] "college"    "police"      "congress"    "thank"       "service"
 [6,] "workforce"  "department"  "legal"       "discuss"     "thank"
 [7,] "schools"    "communities" "dreamers"    "thanks"      "judge"
 [8,] "young"      "protect"     "immigrants"  "meet"        "award"
 [9,] "training"   "officers"    "policy"      "met"         "supreme"
[10,] "future"     "local"       "laws"        "association" "ceremony"
[11,] "career"     "safety"      "dream"       "importance"  "gorsuch"
[12,] "technology" "state"       "dangerous"   "leaders"     "world"
[13,] "help"       "line"        "policies"    "enjoyed"     "justice"
[14,] "student"    "put"         "nation"      "discussion"  "serve"
[15,] "space"      "officer"     "across"      "community"   "vietnam"
      Topic 21         Topic 22        Topic 23    Topic 24      Topic 25
 [1,] "funding"        "rep"           "county"    "house"       "national"
 [2,] "budget"         "members"       "residents" "senate"      "security"
```

```
 [3,] "defense"         "congressman"   "fire"       "vote"        "public"     
 [4,] "infrastructure"  "congress"      "valley"     "floor"       "secretary"  
 [5,] "military"        "joined"        "area"       "colleagues"  "department" 
 [6,] "critical"        "congressional" "center"     "bill"        "social"     
 [7,] "national"        "letter"        "local"      "republicans" "homeland"   
 [8,] "million"         "bipartisan"    "open"       "today"       "park"       
 [9,] "billion"         "caucus"        "management" "republican"  "nation's"   
[10,] "appropriations"  "colleagues"    "please"     "voted"       "border"     
[11,] "important"       "ryan"          "fort"       "pass"        "top"        
[12,] "needs"           "paul"          "evacuation" "now"         "lands"      
[13,] "programs"        "sexual"        "st"         "resolution"  "alaska"     
[14,] "project"         "read"          "stay"       "spoke"       "western"    
[15,] "provides"        "fellow"        "areas"      "democrats"   "important"  
      Topic 26      Topic 27        Topic 28          Topic 29   Topic 30     
 [1,] "federal"     "jobs"          "school"          "live"     "communities"
 [2,] "government"  "businesses"    "high"            "watch"    "water"      
 [3,] "congress"    "business"      "students"        "morning"  "local"      
 [4,] "regulations" "small"         "congressional"   "news"     "working"    
 [5,] "agencies"    "economy"       "district"        "join"     "across"     
 [6,] "spending"    "economic"      "service"         "tonight"  "state"      
 [7,] "rule"        "create"        "academy"         "talk"     "food"       
 [8,] "dollars"     "job"           "young"           "commerce" "farmers"    
 [9,] "wall"        "workers"       "learn"           "see"      "industry"   
[10,] "without"     "growth"        "year"            "discuss"  "agriculture"
[11,] "real"        "help"          "competition"     "tune"     "central"    
[12,] "use"         "development"   "u.s"             "facebook" "like"       
[13,] "debt"        "opportunities" "interested"      "chamber"  "farm"       
[14,] "regulatory"  "grow"          "congratulations" "tomorrow" "rural"      
[15,] "money"       "local"         "app"             "page"     "great"      
      Topic 31      Topic 32      Topic 33   Topic 34   Topic 35   
 [1,] "make"        "act"         "tax"      "family"   "day"      
 [2,] "can"         "bill"        "reform"   "great"    "women"    
 [3,] "work"        "legislation" "families" "time"     "men"      
 [4,] "sure"        "house"       "cuts"     "happy"    "today"    
 [5,] "together"    "passed"      "working"  "everyone" "every"    
 [6,] "need"        "bipartisan"  "taxes"    "day"      "honor"    
 [7,] "better"      "introduced"  "plan"     "many"     "thank"    
 [8,] "making"      "financial"   "code"     "friends"  "nation"   
 [9,] "ways"        "process"     "middle"   "today"    "lives"    
[10,] "come"        "h.r"         "class"    "thanks"   "country"  
[11,] "way"         "protection"  "pay"      "birthday" "remember" 
[12,] "opportunity" "bills"       "jobs"     "hope"     "lost"     
[13,] "bring"       "use"         "bill"     "best"     "sacrifice"
```

```
[14,] "challenges"  "today"       "americans" "celebrate" "brave"
[15,] "families"    "including"   "means"     "wish"      "service"
      Topic 36     Topic 37 Topic 38    Topic 39    Topic 40      Topic 41
 [1,] "health"     "years"  "work"      "american"  "free"        "first"
 [2,] "care"       "year"   "important" "people"    "protect"     "violence"
 [3,] "affordable" "one"    "hard"      "better"    "information" "lives"
 [4,] "act"        "still"  "first"     "made"      "ensure"      "thoughts"
 [5,] "insurance"  "ago"    "step"      "deserve"   "internet"    "families"
 [6,] "healthcare" "two"    "continue"  "time"      "public"      "victims"
 [7,] "repeal"     "time"   "done"      "back"      "must"        "prayers"
 [8,] "access"     "next"   "efforts"   "way"       "rules"       "gun"
 [9,] "obamacare"  "just"   "team"      "promise"   "companies"   "attack"
[10,] "system"     "since"  "part"      "americans" "control"     "steve"
[11,] "costs"      "old"    "across"    "nothing"   "access"      "congress"
[12,] "improve"    "today"  "go"        "put"       "clear"       "taken"
[13,] "quality"    "past"   "hope"      "much"      "protections" "shooting"
[14,] "plans"      "nearly" "can"       "deliver"   "personal"    "responders"
[15,] "premiums"   "days"   "ensuring"  "promises"  "consumers"   "action"
      Topic 42      Topic 43    Topic 44    Topic 45      Topic 46
 [1,] "community"   "america"   "support"   "forward"     "new"
 [2,] "city"        "must"      "proud"     "look"        "u.s"
 [3,] "texas"       "country"   "continue"  "working"     "air"
 [4,] "dr"          "us"        "i'm"       "congress"    "force"
 [5,] "university"  "world"     "fight"     "looking"     "york"
 [6,] "center"      "nation"    "research"  "strong"      "mexico"
 [7,] "de"          "stand"     "cancer"    "move"        "support"
 [8,] "san"         "come"      "fighting"  "committed"   "ensure"
 [9,] "la"          "always"    "stand"     "work"        "guard"
[10,] "el"          "together"  "keep"      "opportunity" "facility"
[11,] "kansas"      "around"    "make"      "priorities"  "coast"
[12,] "king"        "across"    "risk"      "i'm"         "base"
[13,] "congressman" "community" "disease"   "many"        "state's"
[14,] "jr"          "threats"   "awareness" "continuing"  "jersey"
[15,] "building"    "hate"      "advocates" "serve"       "representatives"
      Topic 47      Topic 48     Topic 49     Topic 50
 [1,] "hall"        "can"        "veterans"   "just"
 [2,] "questions"   "today"      "va"         "get"
 [3,] "town"        "help"       "service"    "it's"
 [4,] "join"        "get"        "care"       "one"
 [5,] "event"       "open"       "ensure"     "going"
 [6,] "staff"       "now"        "veteran"    "right"
 [7,] "hope"        "find"       "affairs"    "people"
 [8,] "please"      "plan"       "benefits"   "now"
```

```
 [9,] "office"       "visit"      "department" "time"
[10,] "may"          "sign"       "services"   "good"
[11,] "hosting"      "family"     "medical"    "stop"
[12,] "constituents" "days"       "receive"    "think"
[13,] "call"         "less"       "access"     "back"
[14,] "hours"        "enrollment" "choice"     "another"
[15,] "county"       "deadline"   "served"     "i'm"
```

```
# get_terms(mylda, 15)[, c(3, 4, 16, 30, 36)]

# To get a more granular view, extract the probabilities
mylda_posterior <- topicmodels::posterior(object = mylda)
topic_distr_over_words <- mylda_posterior$terms
topic_distr_over_words_dt <- data.table(topic=1:K,
                                        topic_distr_over_words)
topic_distr_over_words_dt <- melt.data.table(topic_distr_over_words_dt, # If you use base R,
                                             id.vars = 'topic')


# Tidyr/Dplyr way of extracting top 15 rows by group
top10per_topic <- topic_distr_over_words_dt %>%
  group_by(topic) %>%
  slice_max(order_by = value, n = 15)

# data.table way of extracting top 10 rows by group
topic_distr_over_words_dt <- topic_distr_over_words_dt[order(value,decreasing = T)]
top10per_topic <- topic_distr_over_words_dt[,head(.SD,10),by='topic']

topic_name <- c("national security", "drug", "education", "agriculture", "healthcare system")
# Plot probability over words for a few topics.
ggplot(top10per_topic[topic %in% c(3, 4, 16, 30, 36)],aes(y=factor(variable),x=value)) +
  geom_bar(stat = 'identity', position = 'dodge') +
  facet_wrap(~factor(topic,
                     levels = c(3, 4, 16, 30, 36),
                     labels = c("national security", "drug", "education", "agriculture", "hea
                     ),
             scales = 'free')
```



After examining the top 15 words for all topics, I selected topics 3, 4, 16, 30, and 36 and provided appropriate labels for each.

Overall, most topics are clear and meaningful, but there are some "junk" topics. For example, topic 50 do not convey any meaningful information and mainly consist of irrelevant words.

5

```
# To validate labeling; identify documents with highest proportion on any given topic
doc_topic_proportions <- mylda_posterior$topics
# dim(doc_topic_proportions)
# head(doc_topic_proportions)
doc_topic_proportions_dt <- data.table(doc_id = mylda@documents,
                                       doc_topic_proportions)
# here I choose topic 3 and topic 36

colnames(doc_topic_proportions_dt)[2:ncol(doc_topic_proportions_dt)] <- paste0('Topic',colnam

topic3doc <- doc_topic_proportions_dt[order(Topic3,decreasing = T)][1:3,]
topic36doc <- doc_topic_proportions_dt[order(Topic36,decreasing = T)][1:3,]
```

```r
# Select rows from original data...
# national security
cat("Top 3 documents for topic 3 are:", topic3doc$doc_id)
```

```
Top 3 documents for topic 3 are: 4678 4201 2176
```

```r
# fb_congress_data[doc_id %in% topic3doc$doc_id,]$message

# obamacare
cat("Top 3 documents for topic 36 are:", topic36doc$doc_id)
```

```
Top 3 documents for topic 36 are: 2405 4501 489
```

```r
# fb_congress_data[doc_id %in% topic36doc$doc_id,]$message
```

From the topics I labeled, I chose topic 3 and topic 36, which correspond to "national security" and "healthcare system", respectively.

For each of these two topics, I reported the three documents with the highest topic proportions. After reading them qualitatively, I found that they indeed support the themes I assigned.

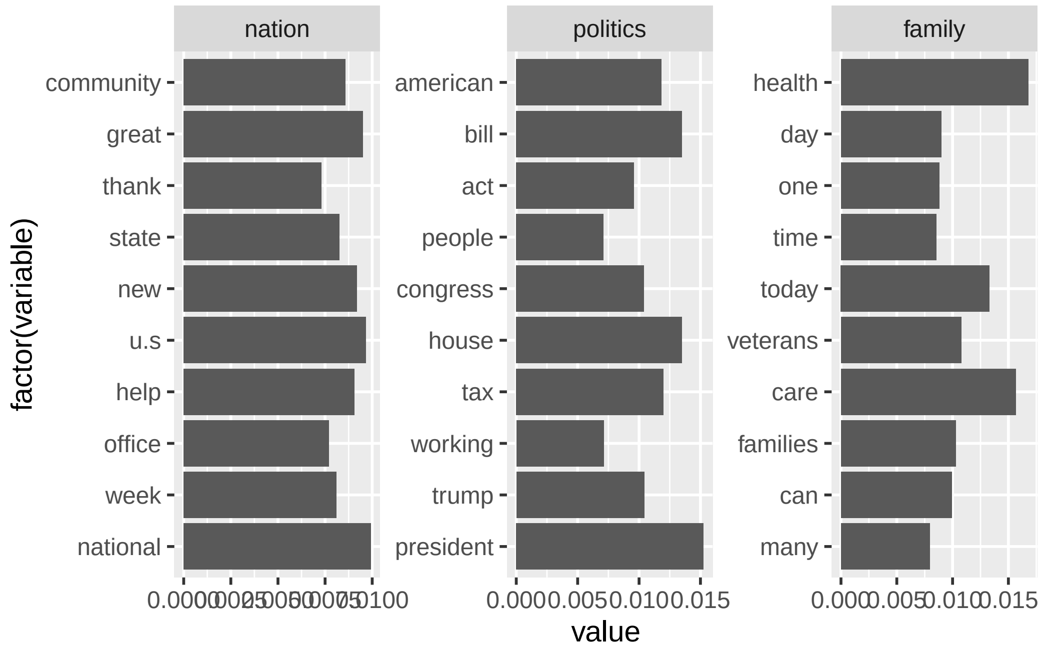
6.

```r
set.seed(1)
K <- 3
mylda_3 <- LDA(x = dtm,
             k = K,
             method="Gibbs",
             control=list(iter = 1000,
                          seed = 1,
                          verbose = 0))

# Inspect top 10 from each topics
get_terms(mylda_3, 10)
```

```
      Topic 1    Topic 2     Topic 3
 [1,] "national" "president" "health"
 [2,] "u.s"      "house"     "care"
```

```
 [3,] "great"     "bill"      "today"
 [4,] "new"       "tax"       "veterans"
 [5,] "help"      "american"  "families"
 [6,] "community" "trump"     "can"
 [7,] "state"     "congress"  "day"
 [8,] "week"      "act"       "one"
 [9,] "office"    "working"   "time"
[10,] "thank"     "people"    "many"
```

```r
# To get a more granular view, extract the probabilities
mylda_3_posterior <- topicmodels::posterior(object = mylda_3)
topic_distr_over_words <- mylda_3_posterior$terms
topic_distr_over_words_dt <- data.table(topic=1:K,
                                        topic_distr_over_words)
topic_distr_over_words_dt <- melt.data.table(topic_distr_over_words_dt, # If you use base R,
                                             id.vars = 'topic')



# Tidyr/Dplyr way of extracting top 15 rows by group
top10per_topic <- topic_distr_over_words_dt %>%
  group_by(topic) %>%
  slice_max(order_by = value, n = 15)

# data.table way of extracting top 10 rows by group
topic_distr_over_words_dt <- topic_distr_over_words_dt[order(value,decreasing = T)]
top10per_topic <- topic_distr_over_words_dt[,head(.SD,10),by='topic']

# Plot probability over words for a few topics.
ggplot(top10per_topic[topic %in% 1:3],aes(y=factor(variable),x=value)) +
  geom_bar(stat = 'identity', position = 'dodge') +
  facet_wrap(~factor(topic,
                     labels = c("nation", "politics", "family")
                     ),
             scales = 'free')
```




```
# To validate labeling; identify documents with highest proportion on any given topic
doc_topic_proportions <- mylda_3_posterior$topics
dim(doc_topic_proportions)
```

```
[1] 5748    3
```

```
head(doc_topic_proportions)
```

```
          1         2         3
1 0.2979798 0.4191919 0.2828283
2 0.3777778 0.2944444 0.3277778
3 0.2348485 0.4393939 0.3257576
4 0.4278607 0.2935323 0.2786070
5 0.2635659 0.4031008 0.3333333
6 0.2666667 0.4196078 0.3137255
```

```
doc_topic_proportions_dt <- data.table(doc_id = mylda_3@documents,
                                       doc_topic_proportions)
colnames(doc_topic_proportions_dt)[2:ncol(doc_topic_proportions_dt)] <- paste0('Topic',colnam

topic2doc <- doc_topic_proportions_dt[order(Topic2,decreasing = T)][1:3,]
```

```r
topic3doc <- doc_topic_proportions_dt[order(Topic3,decreasing = T)][1:3,]

cat("Top 3 documents for topic 2 are:", topic2doc$doc_id)
```

```
Top 3 documents for topic 2 are: 1744 2443 1899
```

```r
cat("Top 3 documents for topic 3 are:", topic3doc$doc_id)
```

```
Top 3 documents for topic 3 are: 4108 290 857
```

```r
# Select rows from original data...
# fb_congress_data[doc_id==1744,]$message
# fb_congress_data[doc_id==4108,]$message
```

Overall, the new model (with $K = 3$) does not capture the topics very well.

I prefer $K = 50$ over $K = 3$ because having only three topics is too coarse and cannot capture the variety and nuances of the different topics discussed.

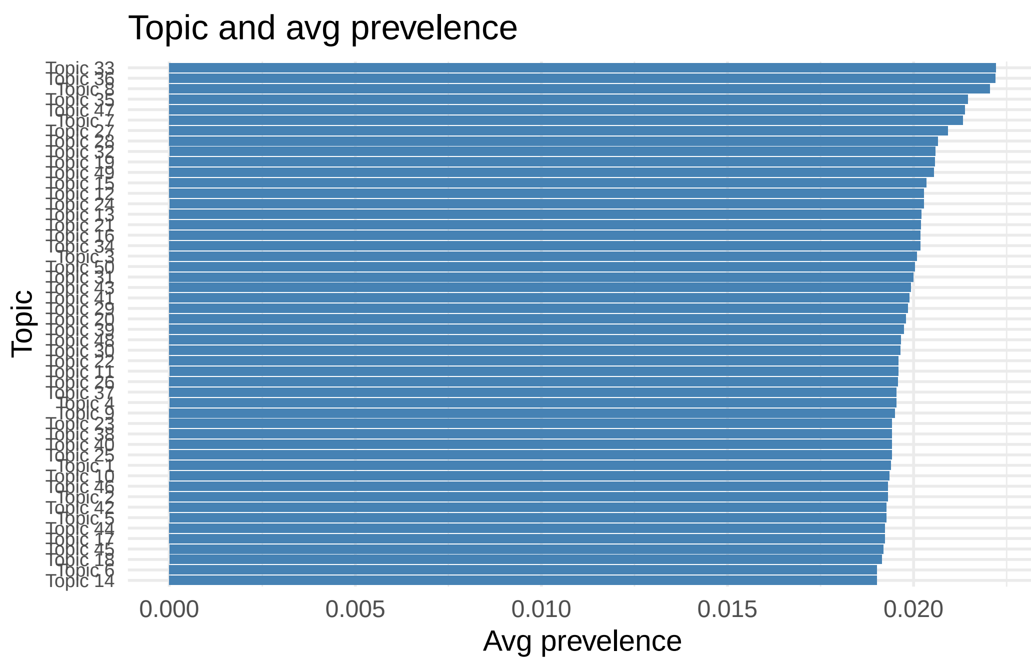
7.

```r
posterior_results <- posterior(mylda)

theta <- posterior_results$topics

topic_prevalence <- colMeans(theta)


cat("The most prevalence is topic", which.max(topic_prevalence))
```

```
The most prevalence is topic 33
```

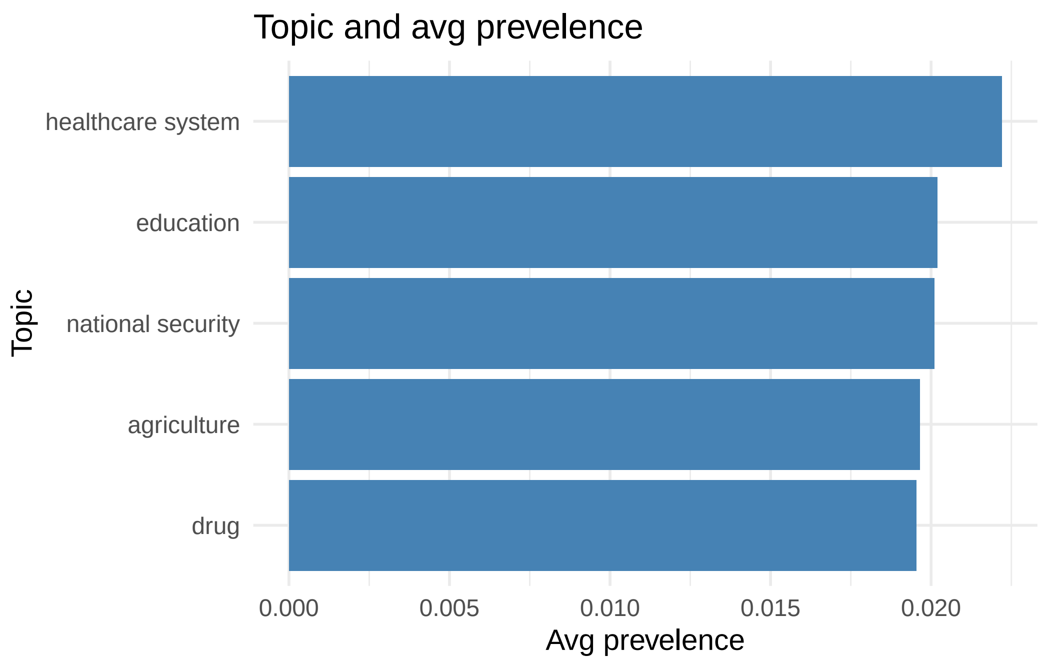
```r
labeled_topics <- c(1:length(topic_prevalence))

plot_data <- data.frame(
  topic = paste0("Topic ", labeled_topics),
  prevalence = topic_prevalence[labeled_topics]
)
```

```r
ggplot(plot_data, aes(x = reorder(topic, prevalence), y = prevalence)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  coord_flip() +
  labs(
    x = "Topic",
    y = "Avg prevelence",
    title = "Topic and avg prevelence"
  ) +
  theme_minimal() +
  theme(axis.text.y = element_text(size = 7))
```




```r
plot_data_subset <- plot_data[c(3, 4, 16, 30, 36), ] |>
  mutate(
    topic = c(3, 4, 16, 30, 36),
    topic_label = factor(topic, levels = c(3, 4, 16, 30, 36), labels = topic_name))

ggplot(plot_data_subset, aes(x = reorder(topic_label, prevalence), y = prevalence)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  coord_flip() +
  labs(
    x = "Topic",
```

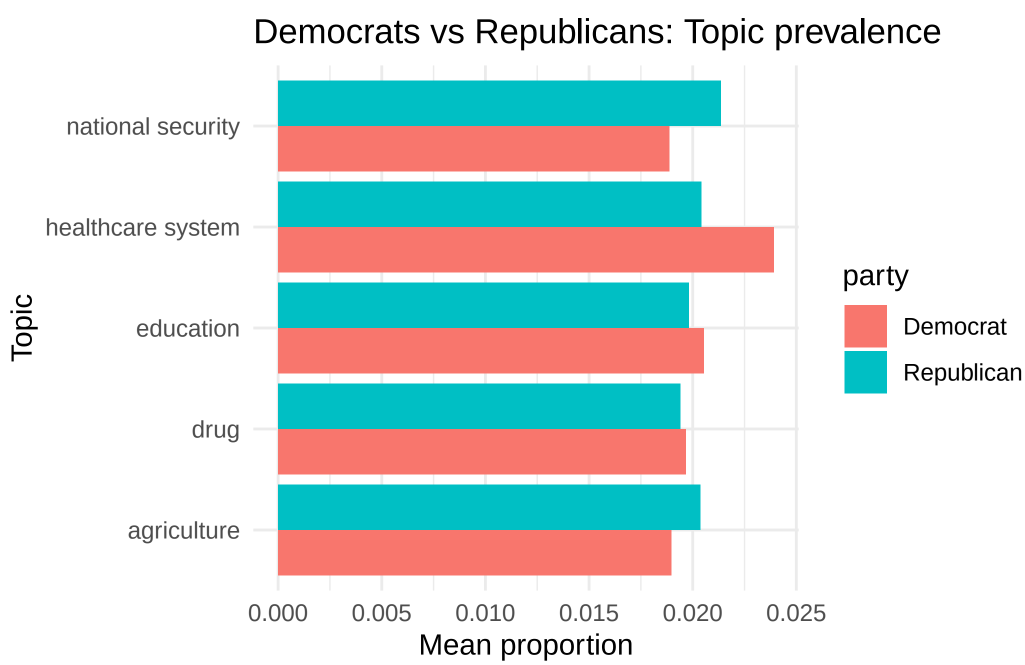
```r
    y = "Avg prevelence",
    title = "Topic and avg prevelence"
  ) +
  theme_minimal()
```



The most common topic is topic 33.

```r
party_raw <- docvars(dtm, "party")

K <- ncol(theta)
colnames(theta) <- paste0("Topic ", 1:K)

theta_df <- as.data.frame(theta) %>%
  mutate(party = party_raw)

theta_long <- theta_df %>%
  pivot_longer(cols = starts_with("Topic"),
               names_to = "topic",
               values_to = "proportion")

labeled_topics <- paste0("Topic ", c(3, 4, 16, 30, 36))
```

```r
theta_long_labeled <- theta_long %>% filter(topic %in% labeled_topics)

mapping <- c(
  "Topic 3"  = "national security",
  "Topic 4"  = "drug",
  "Topic 16" = "education",
  "Topic 30" = "agriculture",
  "Topic 36" = "healthcare system"
)

theta_long_labeled$topic_label <- mapping[theta_long_labeled$topic]

ggplot(theta_long_labeled,
       aes(x = topic_label, y = proportion, fill = party)) +
  stat_summary(fun = mean, geom = "bar", position = "dodge") +
  labs(x = "Topic", y = "Mean proportion",
       title = "Democrats vs Republicans: Topic prevalence") +
  theme_minimal() +
  coord_flip()
```



From the figure above, we can see that there are clear differences in the prevalence of national security, healthcare system, and agriculture topics between Democrats and Republicans.

```r
# glm

glm_topic30 <- glm(`Topic 30` ~ party, family = "quasibinomial", data = theta_df)
# summary(glm_topic30)$coefficients
# broom::tidy(glm_topic30, exponentiate = TRUE)

glm_results <- lapply(labeled_topics, function(tnm) {
  frm <- as.formula(paste0("`", tnm, "` ~ party"))
  fit <- glm(frm, family = "quasibinomial", data = theta_df)
  broom::tidy(fit, exponentiate = TRUE) |> mutate(topic = tnm)
}) |> bind_rows()

glm_wide <- glm_results |>
  select(topic, term, estimate) |>
  pivot_wider(names_from = term, values_from = estimate) |>
  mutate(topic_label = mapping[topic]) |>
  relocate(topic_label, .after = topic)

glm_wide |> kable()
```

topic	topic_label	(Intercept)	partyRepublican
Topic 3	national security	0.0192342	1.1351985
Topic 4	drug	0.0200616	0.9858144
Topic 16	education	0.0209708	0.9644434
Topic 30	agriculture	0.0193341	1.0751342
Topic 36	healthcare system	0.0245023	0.8502475

For topic 3, the odds ratio is 1.135, indicating that Republican members are about 13.5% more likely to post about this topic than Democratic members. In contrast, for topic 36, the odds ratio is 0.850, meaning that Republican members are about 15% less likely to engage with this topic compared to Democrats. The odds ratios for topics 4, 16, and 30 are closer to 1, suggesting that the party differences for these topics are relatively small.

```r
# t test
res <- t.test(theta_df[["Topic 3"]] ~ theta_df$party)
# res$p.value

t_results <- lapply(labeled_topics, function(tnm){
  res <- t.test(theta_df[[tnm]] ~ theta_df$party)
  data.frame(
```

```
    topic = tnm,
    p_value = res$p.value,
    mean_in_Democrat = res$estimate[1],
    mean_in_Republican = res$estimate[2]
  )
}
  ) |>
  bind_rows()
rownames(t_results) <- NULL

t_results <- t_results |>
  mutate(topic_label = mapping[topic]) |>
  relocate(topic_label, .after = topic)

t_results |> kable()
```

topic	topic_label	p_value	mean_in_Democrat	mean_in_Republican
Topic 3	national security	0.0000331	0.0188712	0.0213681
Topic 4	drug	0.5487106	0.0196671	0.0193935
Topic 16	education	0.1754252	0.0205400	0.0198242
Topic 30	agriculture	0.0024109	0.0189674	0.0203634
Topic 36	healthcare system	0.0000001	0.0239163	0.0204079

According to the results of the t-test, topics 3 and 30 are issues that Republicans care more about, while topic 36 is an issue that Democrats care more about. In contrast, there are no significant differences for topics 4 and 16.

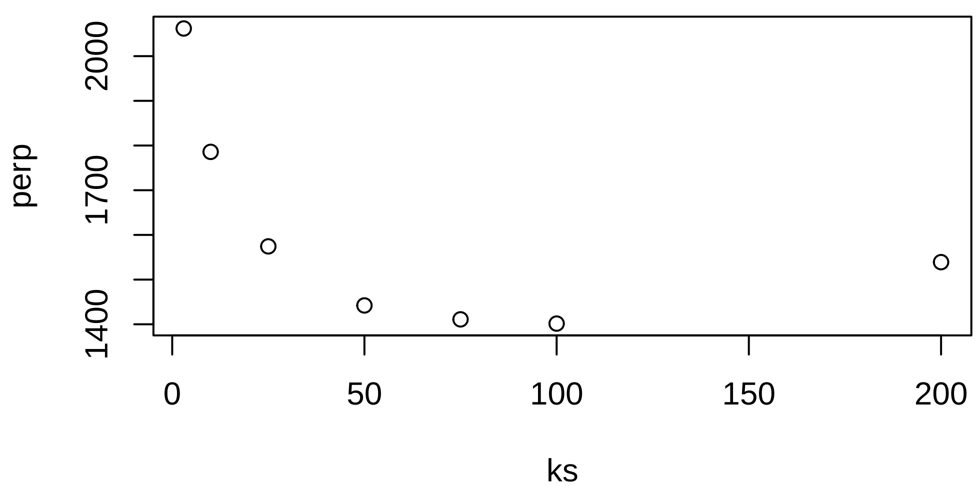
8.

```
set.seed(1984)
row_ids <- 1:nrow(dtm)
train_ids <- sample(x = row_ids, size = 0.8*length(row_ids), replace = F)
test_ids <- row_ids[!row_ids %in% train_ids]
train_dt <- dtm[train_ids,]
test_dt <- dtm[-train_ids,]

ks <- c(3, 10, 25, 50, 75, 100, 200)
perp <- c()
```

```r
for(i in 1:length(ks)){
  current_tm <- LDA(x = train_dt,
                    k = ks[i],
                    method="Gibbs",
                    control=list(iter = 500,
                                 seed = 1,
                                 verbose = 0))
  # Compute perplexity
  perp[i] <- perplexity(object = current_tm,
                        newdata = as.simple_triplet_matrix(test_dt)) # For some reason, perpl
  print(perp)
}
```

```
[1] 2061.894
[1] 2061.894 1785.798
[1] 2061.894 1785.798 1574.329
[1] 2061.894 1785.798 1574.329 1442.176
[1] 2061.894 1785.798 1574.329 1442.176 1411.016
[1] 2061.894 1785.798 1574.329 1442.176 1411.016 1401.556
[1] 2061.894 1785.798 1574.329 1442.176 1411.016 1401.556 1539.033
```

```r
plot(x=ks,y=perp)
```



According to the elbow method, $K = 50$ is the best choice because it is the last point where perplexity drops significantly. Increasing K beyond this point does not lead to a substantial decrease in perplexity. In practice, however, one should also consider which value is more interpretable before making the final choice.

Part 2:

1.

```r
fb_congress_data <- fread("~/Documents/ML/Labs/Lab5/fb-congress-data3.csv")
```

```r
posts_corpus <- corpus(x = fb_congress_data,
                       text_field = "message",
                       meta = list("screen_name", "party"),
                       docid_field = 'doc_id')

# print(posts_corpus)

# Tokenize & clean from particular types of words
mytokens <- tokens(x = posts_corpus,
                   remove_punct = TRUE,
                   remove_numbers = TRUE,
                   remove_symbols = TRUE,
                   remove_url = TRUE,
                   padding = FALSE)
# print(mytokens)

# Make tokens lowercase
mytokens <- tokens_tolower(x = mytokens)

# How does it look now?
# mytokens[1:3]

mysentences <- sapply(mytokens, function(x) paste(x, collapse = " "))
# mysentences[1:3]
```

2.

```
set.seed(1984)
w2v <- word2vec(x = mysentences,          # Your data
                            type = "cbow",    # Model type (the one we talked about in lectu
                            window = 15,       # Context defined in terms of +-5 words.
                            dim = 50,          # Dimensionality of embeddings
                            iter = 50,         # Estimation iterations (higher means more tim
                            hs = FALSE,        # Setting to FALSE here --> "negative sampling
                            negative = 15)   # Number of negative samples (how many fake sen

# w2v_association <- word2vec(x = mysentences,          # Your data
#                             type =  "skip-gram",    # Model type (the one we talked about
#                             window = 15,       # Context defined in terms of +-5 words.
#                             dim = 50,          # Dimensionality of embeddings
#                             iter = 50,         # Estimation iterations (higher means more t
#                             hs = FALSE,        # Setting to FALSE here --> "negative sampli
#                             negative = 15)   # Number of negative samples (how many fake s
```

3.

```
selected_words <- c("russia", "tax", "trump")
w2v_predict <- predict(w2v, selected_words, type = "nearest", top_n = 10)

w2v_predict
```

```
$russia
    term1         term2 similarity rank
1  russia       russian  0.8875819    1
2  russia   connections  0.8408329    2
3  russia      election  0.8207063    3
4  russia      campaign  0.8182974    4
5  russia investigation  0.8106385    5
6  russia    associates  0.8091059    6
7  russia       inquiry  0.8078038    7
8  russia  interference  0.8032950    8
9  russia         putin  0.8018053    9
10 russia     collusion  0.8017161   10

$tax
   term1       term2 similarity rank
```

```
1     tax        taxes  0.8022894    1
2     tax     windfall  0.7324714    2
3     tax    draconian  0.7263859    3
4     tax   simplifies  0.7232710    4
5     tax     standard  0.7087603    5
6     tax    trillions  0.6925077    6
7     tax    taxreform  0.6827784    7
8     tax         nfip  0.6824735    8
9     tax desperately  0.6667807    9
10    tax    paychecks  0.6594531   10

$trump
   term1   term2 similarity rank
1  trump trump's  0.9256474    1
2  trump   obama  0.8930461    2
3  trump obama's  0.8347976    3
4  trump    vice  0.8233221    4
5  trump  reagan  0.7706845    5
6  trump kennedy  0.7569413    6
7  trump    tone  0.7395135    7
8  trump refused  0.7358065    8
9  trump  hiding  0.7308963    9
10 trump   elect  0.7226571   10
```

The prediction results are relatively consistent with expectations. However, in fact, a model trained using the skip-gram approach would produce results that align even more closely with expectations.

4.

```r
embedding <- as.matrix(w2v)

embedding[rownames(embedding)=="women",]
```

```
 [1]  1.45846772 -0.02067013 -0.77867019 -0.26792613 -0.82894021  0.46333125
 [7]  2.39504385 -0.53787041 -1.01073909  0.03637369 -0.75768471  0.92142916
[13]  0.97042996 -0.23480754  1.57735109  0.14422640  0.25721845 -0.56702614
[19] -0.29286489 -1.01375794 -1.29568195  0.37645638  0.84087372 -2.64823031
[25] -0.32585651  0.77929771 -0.78593844  2.06397152  1.28935301 -1.22126913
[31]  0.73119563  1.29744661  1.18119347 -0.35590121  0.22335435  0.05245591
```

```
[37]  0.09045754 -0.26514432  0.18268888 -1.01847148  1.40893888 -1.28125286
[43]  0.19028999  0.01757974 -0.10425572 -0.10621946  0.33540925 -2.30685878
[49]  0.46995500 -0.49623686
```

```r
kingtowoman <- embedding["king", ] - embedding["man", ] + embedding["woman", ]
sims <- word2vec_similarity(embedding, matrix(kingtowoman, nrow = 1))

head(sort(sims[, 1], decreasing = TRUE), 20)
```

```
       king     towards      luther       woman      martin     phoenix
  0.9581525   0.8444773   0.8133649   0.8120883   0.8115314   0.7921137
     strive      legacy        tide        luke achievement     strides
  0.7641762   0.7586740   0.7529641   0.7528000   0.7509814   0.7508735
        eye   holocaust       aside    enduring      degree       child
  0.7508017   0.7502565   0.7426112   0.7377294   0.7323822   0.7318026
    respond  connection
  0.7282208   0.7207185
```

```r
pretrained <- readRDS("~/Documents/ML/Labs/Lab5/glove6B200d.rds")
dim(pretrained)
```

```
[1] 400000    200
```

```r
# embedding_pretained <- as.matrix(pretrained) # pretrained is "matrix" "array"
# pretrained[rownames(pretrained)=="king",]
kingtowoman <- pretrained["king", ] - pretrained["man", ] + pretrained["woman", ]
sims <- word2vec_similarity(pretrained, matrix(kingtowoman, nrow = 1))
head(sort(sims[, 1], decreasing = TRUE), 20)
```

```
     king     queen  princess    prince    throne   emperor     royal  daughter
0.4630444 0.4272459 0.4001166 0.3994429 0.3877838 0.3765438 0.3732692 0.3721567
  monarch   kingdom     crown    mother       her  marriage      wife elizabeth
0.3719335 0.3668355 0.3652451 0.3617875 0.3608811 0.3567054 0.3565992 0.3557918
    woman   duchess adulyadej      duke
0.3553641 0.3549088 0.3543496 0.3536174
```

```r
dist_vec <- function(a,b) sqrt(sum((a-b)^2))
dist_sta_man <- dist_vec(pretrained["statistician", ], pretrained["man", ])
dist_sta_woman <- dist_vec(pretrained["statistician", ], pretrained["woman", ])
occupationalbias <- dist_sta_man - dist_sta_woman
occupationalbias
```

```
[1] -0.3029102
```

The occupational bias is -0.303, indicating that “statistician” is farther from “woman” than from “man” in the embedding space, which suggests the presence of occupational bias.

5.

```r
corpus_dem <- corpus_subset(posts_corpus, party == "Democrat")
corpus_rep <- corpus_subset(posts_corpus, party == "Republican")

mytokens_dem <- tokens(x = corpus_dem,
                   remove_punct = TRUE,
                   remove_numbers = TRUE,
                   remove_symbols = TRUE,
                   remove_url = TRUE,
                   padding = FALSE)

mytokens_rep <- tokens(x = corpus_rep,
                   remove_punct = TRUE,
                   remove_numbers = TRUE,
                   remove_symbols = TRUE,
                   remove_url = TRUE,
                   padding = FALSE)

# Make tokens lowercase
mytokens_dem <- tokens_tolower(x = mytokens_dem)
mytokens_rep <- tokens_tolower(x = mytokens_rep)

mysentences_dem <- sapply(mytokens_dem, function(x) paste(x, collapse = " "))
mysentences_rep <- sapply(mytokens_rep, function(x) paste(x, collapse = " "))
```

```r
set.seed(1984)
w2v_dem <- word2vec(x = mysentences_dem,          # Your data
                          type = "cbow",    # Model type (the one we talked about in lectu
                          window = 15,       # Context defined in terms of +-5 words.
                          dim = 50,         # Dimensionality of embeddings
                          iter = 50,        # Estimation iterations (higher means more time
                          hs = FALSE,       # Setting to FALSE here --> "negative sampling
                          negative = 15)   # Number of negative samples (how many fake sen
```

```r
w2v_rep <- word2vec(x = mysentences_rep,          # Your data
                            type = "cbow",    # Model type (the one we talked about in lectu
                            window = 15,       # Context defined in terms of +-5 words.
                            dim = 50,          # Dimensionality of embeddings
                            iter = 50,         # Estimation iterations (higher means more tim
                            hs = FALSE,        # Setting to FALSE here --> "negative sampling
                            negative = 15)   # Number of negative samples (how many fake sen
```

```r
predict(w2v_dem, c("immigrants", "abortion"), type = "nearest", top_n = 10)
```

```
$immigrants
        term1       term2 similarity rank
1  immigrants    refugees  0.7158788    1
2  immigrants solidarity  0.7004563    2
3  immigrants    dreamers  0.6997086    3
4  immigrants    profound  0.6945156    4
5  immigrants     sisters  0.6927473    5
6  immigrants   immigrant  0.6916040    6
7  immigrants     granted  0.6829141    7
8  immigrants   religious  0.6797104    8
9  immigrants      values  0.6790193    9
10 immigrants     circuit  0.6628950   10

$abortion
      term1         term2 similarity rank
1  abortion controversial  0.7209508    1
2  abortion    prevention  0.7096908    2
3  abortion  reproductive  0.7078452    3
4  abortion       prevent  0.6851029    4
5  abortion  restrictions  0.6730043    5
6  abortion contraception  0.6706381    6
7  abortion       control  0.6562243    7
8  abortion       missile  0.6389863    8
9  abortion      protects  0.6349660    9
10 abortion         human  0.6342913   10
```

```r
predict(w2v_rep, c("immigrants", "abortion"), type = "nearest", top_n = 10)
```

```
$immigrants
        term1     term2 similarity rank
```

```
1  immigrants    crimes  0.8012899    1
2  immigrants sanctuary  0.7992841    2
3  immigrants    aliens  0.7928141    3
4  immigrants  deported  0.7918141    4
5  immigrants dangerous  0.7917460    5
6  immigrants   illegal  0.7888762    6
7  immigrants criminals  0.7773612    7
8  immigrants     crime  0.7529935    8
9  immigrants   violent  0.7480202    9
10 immigrants  criminal  0.7310147   10

$abortion
      term1      term2 similarity rank
1  abortion     unborn  0.8258076    1
2  abortion parenthood  0.7576534    2
3  abortion    planned  0.7444595    3
4  abortion  pregnancy  0.7365335    4
5  abortion    sponsor  0.6795615    5
6  abortion     babies  0.6746088    6
7  abortion          x  0.6689040    7
8  abortion  supported  0.6666662    8
9  abortion  cosponsor  0.6636977    9
10 abortion  heartbeat  0.6615708   10
```

The results are consistent with expectations. For Republicans, the nearest words to "immigrants" are more negative, while for Democrats, the nearest words are relatively more neutral or positive. The same pattern is observed for "abortion", which reflects the highly conservative beliefs of the Republican Party.

6.

```r
positive <- fread("~/Documents/ML/Labs/Lab5/positive.txt")[[1]]
negative <- fread("~/Documents/ML/Labs/Lab5/negative.txt")[[1]]
selected_words <- c("russia", "tax", "trump")
```

```r
embeddings_dem <- as.matrix(w2v_dem)
pos_in_dem <- intersect(positive, rownames(embeddings_dem))
neg_in_dem <- intersect(negative, rownames(embeddings_dem))
pos_matrix_dem <- embeddings_dem[pos_in_dem, , drop = FALSE]
neg_matrix_dem <- embeddings_dem[neg_in_dem, , drop = FALSE]
```

```r
pos_vec_dem <- colMeans(pos_matrix_dem)
neg_vec_dem <- colMeans(neg_matrix_dem)

sentiment_vec_dem <- neg_vec_dem - pos_vec_dem
sims_dem <- word2vec_similarity(embeddings_dem, sentiment_vec_dem)
similarity_table_dem <- data.frame(
  word = rownames(embeddings_dem),
  sim = as.numeric(sims_dem[, 1])
)
sims_dem_filtered <- similarity_table_dem |>
  filter(!(word %in% negative)) |>
  arrange(desc(sim)) |>
  mutate(rank = row_number())
```

```r
embeddings_rep <- as.matrix(w2v_rep)
pos_in_rep <- intersect(positive, rownames(embeddings_rep))
neg_in_rep <- intersect(negative, rownames(embeddings_rep))
pos_matrix_rep <- embeddings_rep[pos_in_rep, , drop = FALSE]
neg_matrix_rep <- embeddings_rep[neg_in_rep, , drop = FALSE]
pos_vec_rep <- colMeans(pos_matrix_rep)
neg_vec_rep <- colMeans(neg_matrix_rep)

sentiment_vec_rep <- neg_vec_rep - pos_vec_rep
sims_rep <- word2vec_similarity(embeddings_rep, sentiment_vec_rep)
similarity_table_rep <- data.frame(
  word = rownames(embeddings_rep),
  sim = as.numeric(sims_rep[, 1])
)
sims_rep_filtered <- similarity_table_rep |>
  filter(!(word %in% negative)) |>
  arrange(desc(sim)) |>
  mutate(rank = row_number())
```

```r
sims_dem_filtered |> head(10)
```

```
         word       sim rank
1      highly 0.3809679    1
2         cbo 0.3594549    2
3      result 0.3561186    3
4   muslimban 0.3523661    4
5      likely 0.3493890    5
```

```
6  associated 0.3489818    6
7      ruling 0.3488382    7
8     attempt 0.3475912    8
9  threatened 0.3460561    9
10   withdraw 0.3433446   10
```

```
sims_rep_filtered |> head(10)
```

```
        word       sim rank
1    massive 0.3800281    1
2    deficit 0.3783081    2
3    certain 0.3738332    3
4     crimes 0.3675560    4
5  destroyed 0.3638638    5
6  wildfires 0.3623684    6
7     spread 0.3613781    7
8     accept 0.3557568    8
9  sanctions 0.3557266    9
10        al 0.3555688   10
```

```
rank_dem <- sims_dem_filtered |> filter(word %in% selected_words)

rank_rep <- sims_rep_filtered |> filter(word %in% selected_words)

rank_dem |>
  kable(caption = "Word Rank for Democrats")
```

Table 3: Word Rank for Democrats

word	sim	rank
trump	0.2818902	138
russia	0.2153687	562
tax	0.1820681	829

```
rank_rep |>
  kable(caption = "Word Rank for Republicans")
```

Table 4: Word Rank for Republicans

word	sim	rank
russia	0.2943141	133
tax	0.0790369	1600
trump	0.0000000	2120

I selected trump, russia, gun, and tax. For example, we can observe that trump has a lower similarity score on the negative sentiment dimension for Republicans, meaning that trump is relatively less negative. In contrast, trump has a higher similarity score on the negative sentiment dimension for Democrats, indicating that trump is perceived as highly negative.

7.

```
fit_and_rank <- function(mysentences, positive_words = postive, negative_words = negative, ty
  w2v <- word2vec(x = mysentences, type = type, window = window, dim = dim,
                           iter = iter, hs = hs, negative = negative)
  emb <- as.matrix(w2v)
  pos_in <- intersect(positive_words, rownames(emb))
  neg_in <- intersect(negative_words, rownames(emb))
  pos_vec <- colMeans(emb[pos_in, , drop = FALSE])
  neg_vec <- colMeans(emb[neg_in, , drop = FALSE])
  sentiment_vec <- neg_vec - pos_vec
  sims <- word2vec_similarity(emb, sentiment_vec)
  similarity_table <- data.frame(
  word = rownames(emb),
  sim = as.numeric(sims[, 1])
)
sims_filtered <- similarity_table |>
  filter(!(word %in% negative_words)) |>
  arrange(desc(sim)) |>
  mutate(rank = row_number())
}

bootstrap_by_party <- function(mysentences, positive_words = positive, negative_words = negat
  set.seed(seed)
  map_dfr(seq_len(n_boot), function(bi) {
    idx <- sample.int(length(mysentences), replace = TRUE)
    sent_boot <- mysentences[idx]
    fit_and_rank(sent_boot, positive_words = positive, negative_words = negative) |>
```

```
      filter(word %in% selected_words) |>
      mutate(iter = bi)
  })
}
```

```
boot_dem <- bootstrap_by_party(mysentences_dem) |>
  mutate(party = "Democrat")
boot_rep <- bootstrap_by_party(mysentences_rep) |>
  mutate(party = "Republican")

boot_all <- bind_rows(boot_dem, boot_rep)

# saveRDS(boot_all, file = "boot_all.rds")

boot_all <- readRDS("boot_all.rds")


boot_summary <- boot_all |>
  group_by(party, word) |>
  summarize(n_iter = n(),
            sim_mean = mean(sim, na.rm = TRUE),
            sim_low = quantile(sim, 0.025, na.rm = TRUE),
            sim_high = quantile(sim, 0.975, na.rm = TRUE),
            rank_mean = mean(rank, na.rm = TRUE),
            .groups = "drop") |>
  mutate(party = recode(party,
                        "Republican" = "rep",
                        "Democrat" = "dem"))

boot_sim_wide <- boot_summary |>
  select(party, word, sim_mean, sim_low, sim_high) |>
  pivot_wider(names_from = party, values_from = c(sim_mean, sim_low, sim_high)) |>
  select(word,
         sim_mean_dem, sim_low_dem, sim_high_dem,
         sim_mean_rep, sim_low_rep, sim_high_rep)

boot_summary |> kable()
```

party	word	n_iter	sim_mean	sim_low	sim_high	rank_mean
dem	gun	20	0.1851828	0.1039368	0.2512373	599.300
dem	russia	20	0.1708284	0.0800868	0.2205586	705.350

party	word	n_iter	sim_mean	sim_low	sim_high	rank_mean
dem	tax	20	0.1565825	0.1163379	0.1971483	841.800
dem	trump	20	0.2342240	0.1938609	0.2666179	210.450
rep	gun	14	0.1022162	0.0000000	0.2428588	1807.643
rep	russia	20	0.2742442	0.2297856	0.3419737	96.700
rep	tax	20	0.0446886	0.0000000	0.1326030	1800.150
rep	trump	20	0.0294843	0.0000000	0.1167477	2032.200

```
boot_sim_wide |>
  kable(
    col.names = c("Word", "Mean (Dem)", "Low (Dem)", "High (Dem)",
                  "Mean (Rep)", "Low (Rep)", "High (Rep)"),
    caption = "Bootstrapped Word Similarity for Selected Words Across Parties (Dem = Democra
  )
```

Table 6: Bootstrapped Word Similarity for Selected Words Across Parties (Dem = Democrat, Rep = Republican)

Word	Mean (Dem)	Low (Dem)	High (Dem)	Mean (Rep)	Low (Rep)	High (Rep)
gun	0.1851828	0.1039368	0.2512373	0.1022162	0.0000000	0.2428588
russia	0.1708284	0.0800868	0.2205586	0.2742442	0.2297856	0.3419737
tax	0.1565825	0.1163379	0.1971483	0.0446886	0.0000000	0.1326030
trump	0.2342240	0.1938609	0.2666179	0.0294843	0.0000000	0.1167477

The conclusion is consistent with the above, but we can observe that only russia has a lower bound greater than zero. This indicates that russia is the only word with a significant difference between the two parties, while the differences for the other words are not statistically significant.